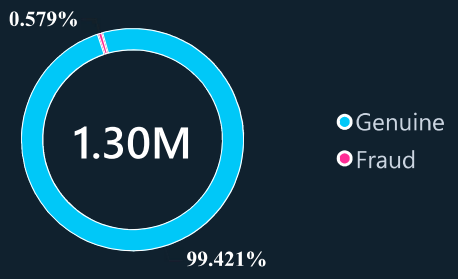
BFSI | Credit Card Fraud Analytics

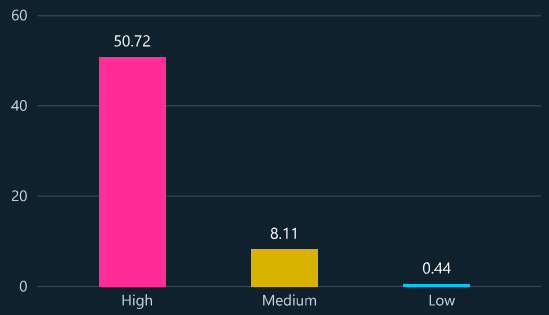
DETECT • ANALYZE • PREVENT • BUILD TRUST

Total Transactions	Fraud Transactions	Fraud Rate (%)	Fraudulent Transaction Value
1.30M	7.51K	0.58%	$3.99M

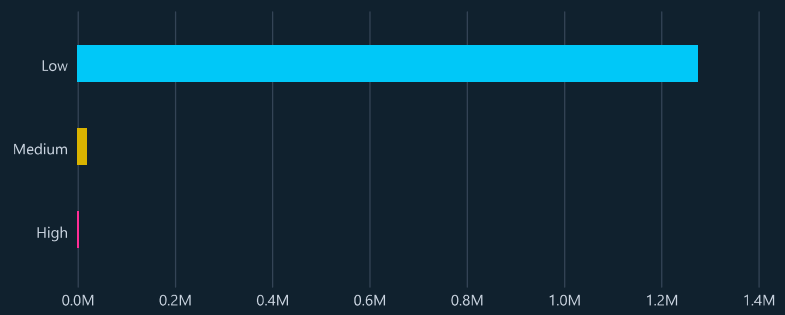
Fraud vs Genuine Transactions



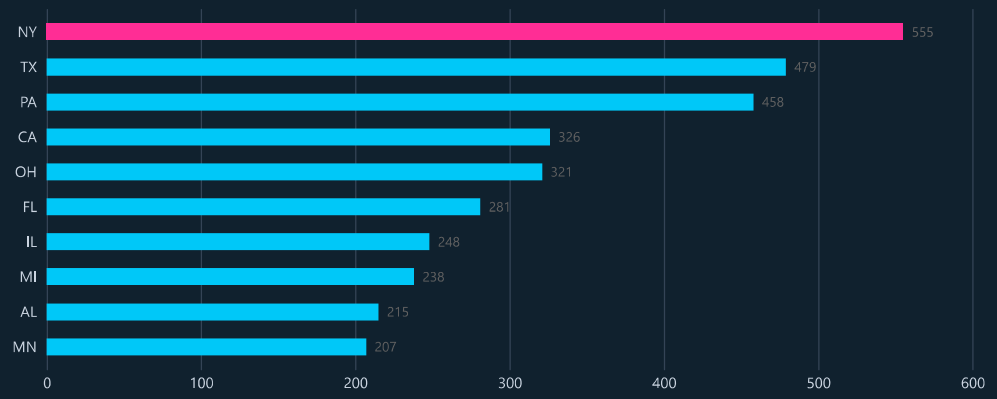
Fraud Rate by Risk Level (%)



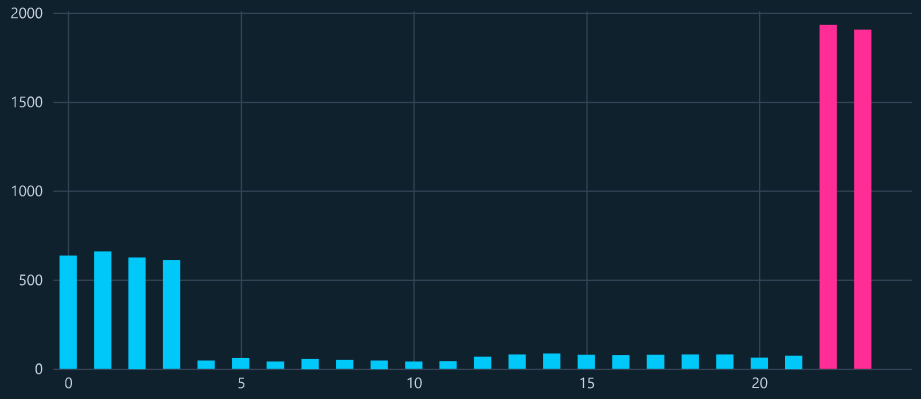
Transactions by Risk Level



Top 10 States by Fraud Transactions



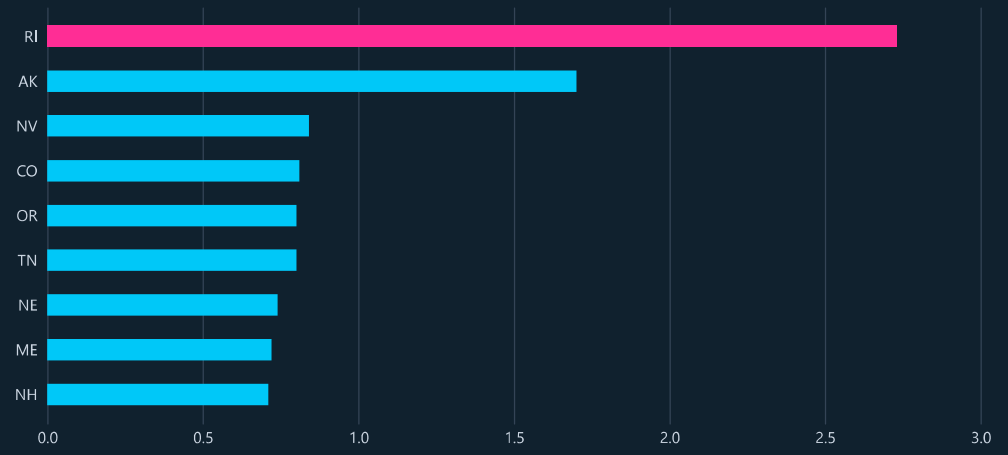
Fraud Transactions by Hour



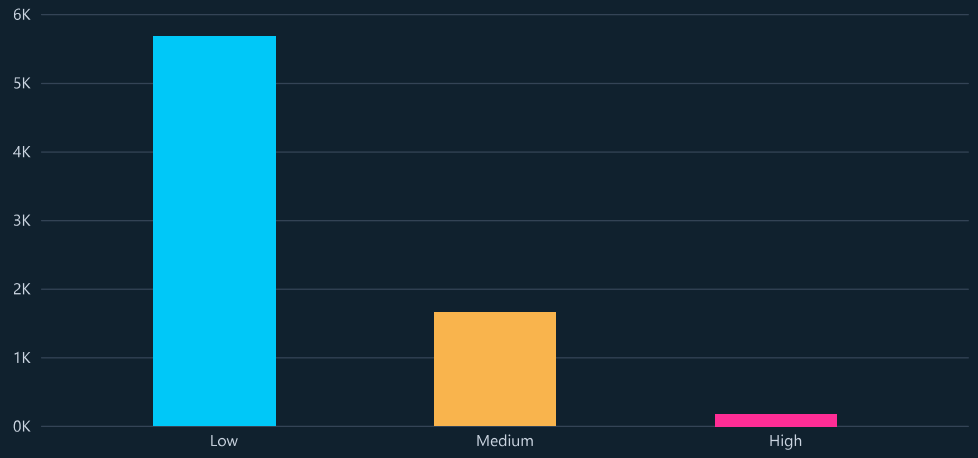
BFSI | Fraud Patterns & Risk Analysis

DISCOVER · ANALYZE · IDENTIFY RISK · PREVENT FRAUD

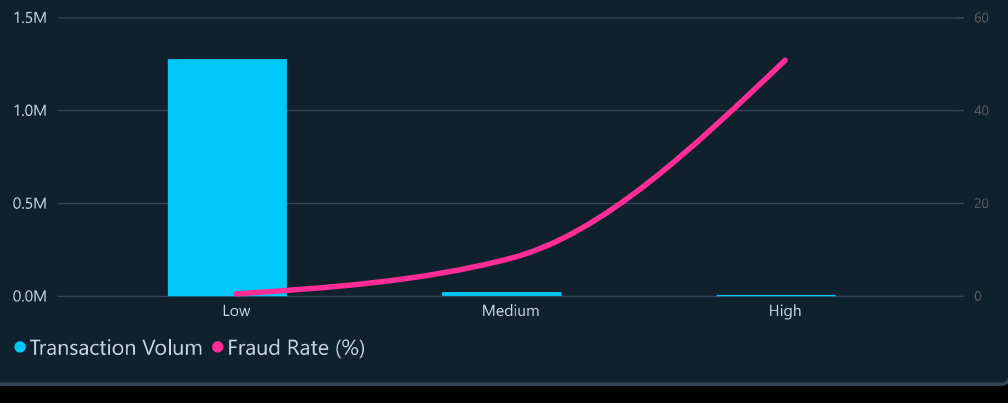
Top 10 States by Fraud Rate (%)



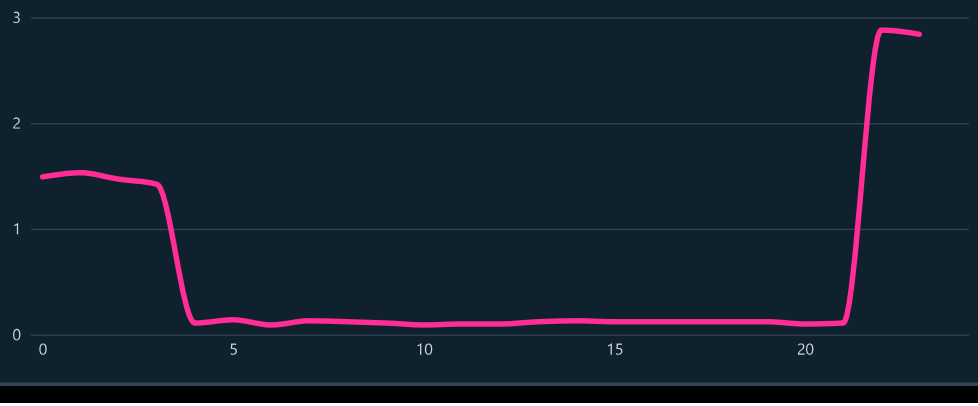
Fraud Transactions by Risk Level



Transaction Volume vs Fraud Rate by Risk Level.



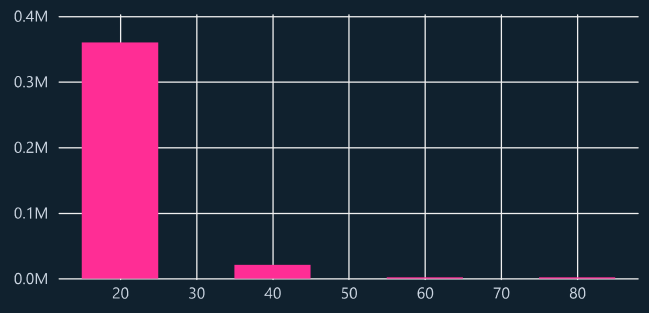
Fraud Rate by Hour (%)



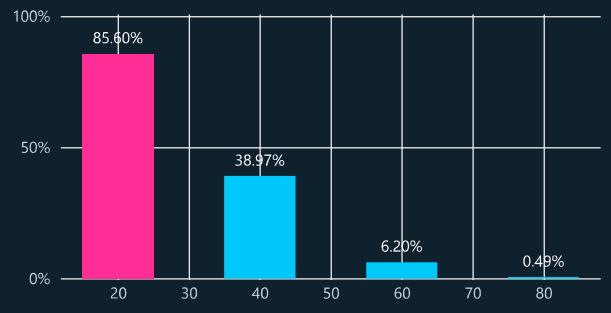
BFSI | Fraud Alert & Risk Performance

MONITOR · EVALUATE · OPTIMIZE · MITIGATE RISK

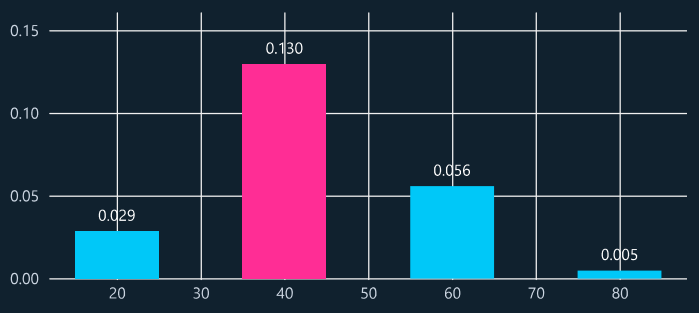
Total Fraud Alerts by Threshold



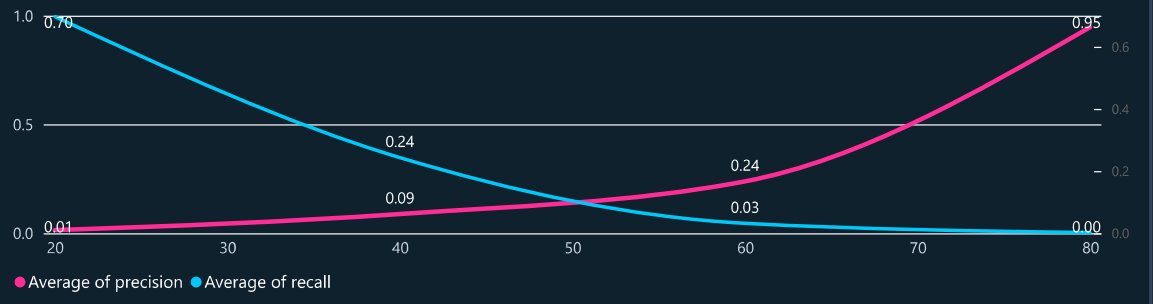
Fraud Value Recall by Alert Threshold



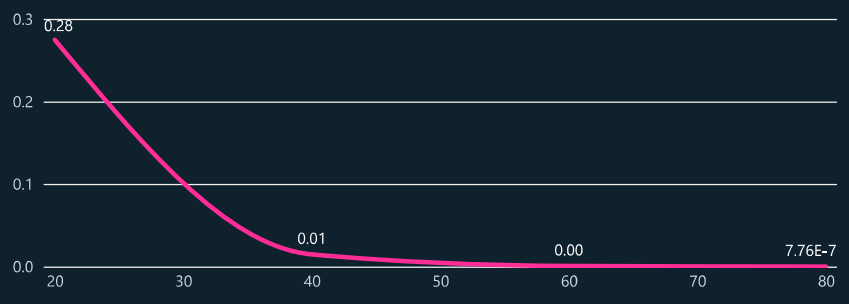
F1 Score by Alert Threshold



Precision vs Recall by Alert Threshold



- False Positive Rate by Alert Threshold



KEY BUSINESS INSIGHTS | FRAUD ALERT PERFORMANCE

Threshold 20: Highest fraud value coverage (85.60%), but generates a large number of alerts.

Threshold 40: Highest F1 Score among tested thresholds (~0.13), offering the strongest observed precision–recall balance.

Threshold 60: Fraud value recall drops to 6.20%, indicating substantial missed fraud value.

Threshold 80: Fraud value recall falls to just 0.49%, making this threshold unsuitable when broad fraud coverage is the priority.

Preliminary Recommendation: Use threshold 40 as the candidate for further validation. Final selection should consider investigation capacity, false-positive costs, and performance on unseen data.